F01 WATERING MANAGMENT

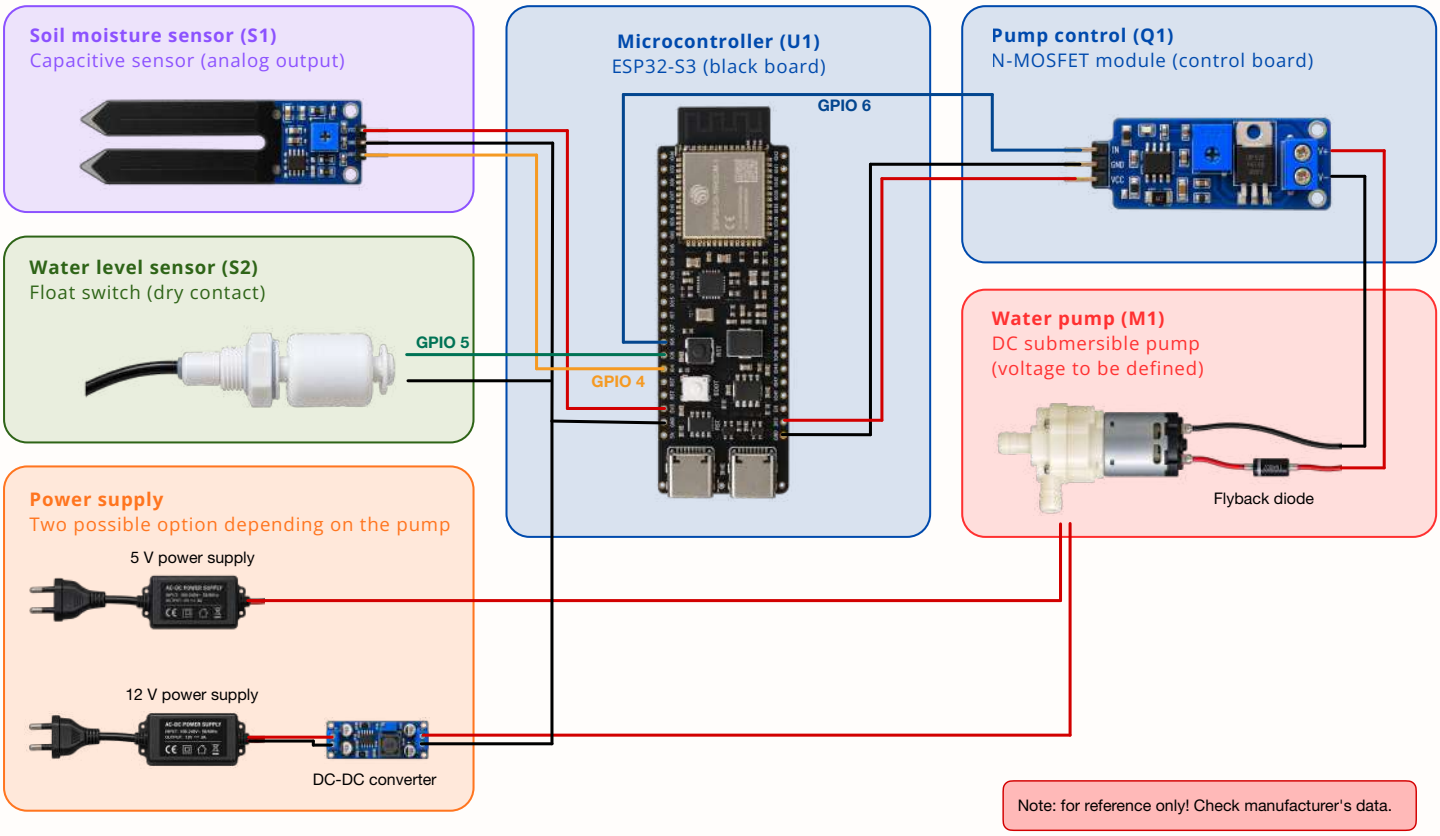
OBJECTIVE

Remotely control the water pump and automatically trigger watering based on the substrate moisture and water level.

MAIN FUNCTION

- Remote control (On / Off)
- Automatic watering based on moisture
- Maximum watering duration
- Empty tank detection (safety)
- Statue monitoring and remote feedback

ELECTRICAL SCHEMATIC



Note: for reference only! Check manufacturer's data.

ESP32 PINOUT TABLE

Function	ESP32-S3 GPIO	Connection	Signal type
Soil moisture	GPIO 4	AOUT sensor S1	Analog (0 - 3.3 V)
Water level	GPIO 5	Float Switch S2	Digital Input
Pump control	GPIO 6	MOSFET Input	Digital (3.3 V)

KEY POINTS

- All ground must be common.
- Use a separate power simple for the pump and ESP32.
- Use a 3.3 V compatible moisture sensor (or a voltage divider if 5 V).
- The flyback diode protect against motor overvoltage.
- Use a fuse adapted to the pump power supply.
- Check component specification before intégration.

COMPONENT LIST

Ref.	Component	Qty	Criteria (see datasheet)
U1	ESP32-S3	1	Development board
S1	Soil moisture sensor	1	Capacitive 3.3 V
S2	Float switch	1	Dry contact
Q1	MOSFET module	1	3.3 V compatible
M1	Water pump	1	5V - 12V (to be defined)
PSU1	Power supply	1	Depending on pump
U2	DC-DC converter	1	12V to 5V (if necessary)

FIRE

FO2 LIGHTING MANAGMENT

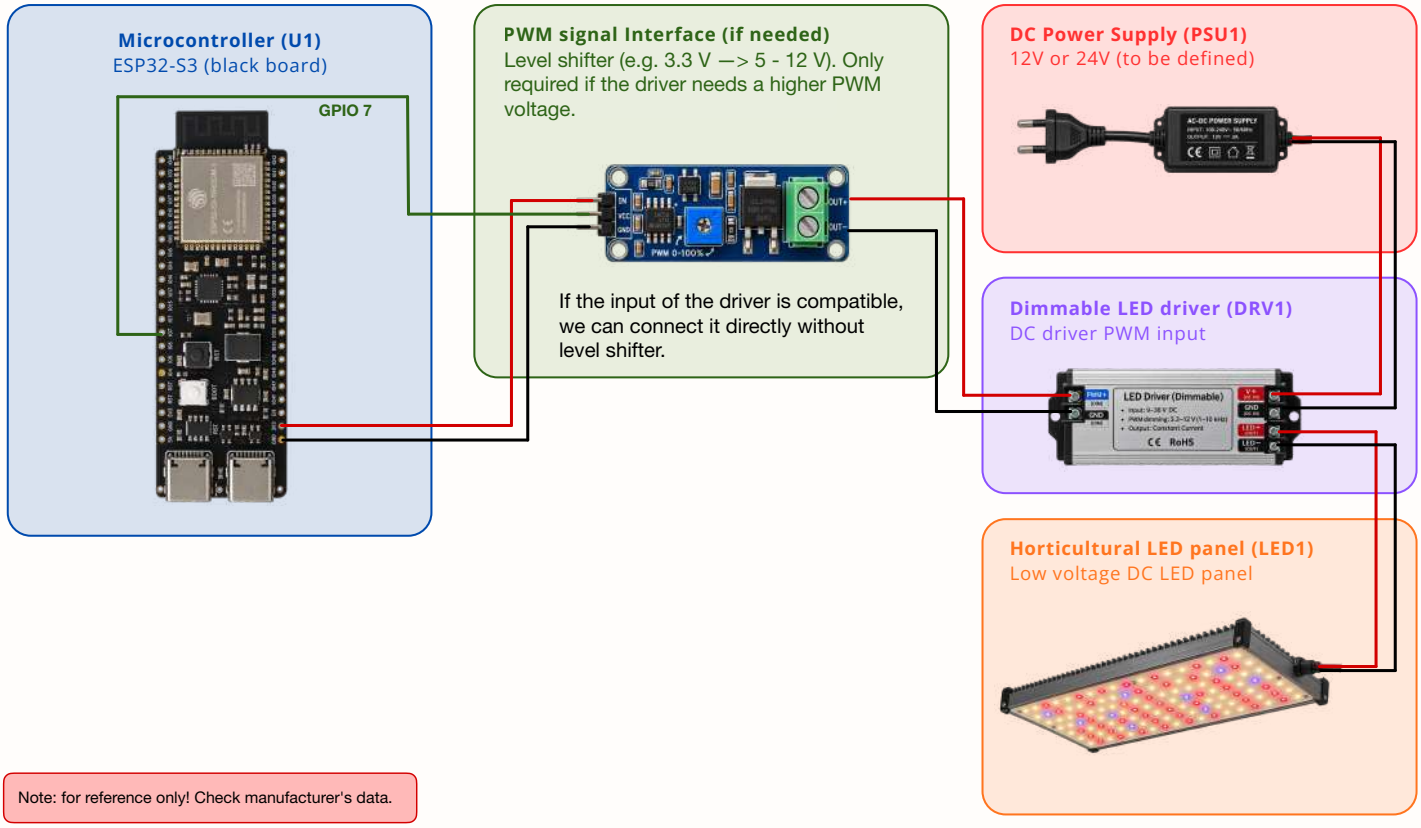
OBJECTIVE

Control the horticultural lighting of the smart garden remotely and adjust its intensity to meet the plants' needs.

MAIN FUNCTION

- Remote control (On / Off)
- Light intensity control (0% - 100%)
- Reliable and safe operation
- Scalable for future feature
- Statue monitoring and remote feedback

ELECTRICAL SCHEMATIC



Note: for reference only! Check manufacturer's data.

ESP32 PINOUT TABLE

Function	ESP32-S3 GPIO	Connection	Signal type
Light control (FO2)	GPIO 7	PWM to LED driver	PWM 3.3 V
Sensors power	3V3	Sensor VCC	3.3 V DC
Common Ground	GND	All ground (logic)	0 V

KEY POINTS

- Check PWM input specification of the led driver.
- Use a level shifter if the driver requires a PWM voltage higher than 3.3 V.
- Ensure all ground are common.
- Choose a driver compatible with the LED panel (voltage and current).
- Consider adding a fuse on the driver power supply.
- Start with a PWM frequency between 100 Hz and 10 kHz.

COMPONENT LIST

Ref.	Component	Qty	Criteria (see datasheet)
U1	ESP32-S3	1	Development board
DRV1	Dimmable LED driver	1	PWM input (3.3 - 12 V, 0.1 - 10 kHz)
LED1	Horticultural LED panel	1	Low voltage DC (eg. 12 - 24 V)
U2	Level shifter (optional)	1	Eg. MOSFET module or transistor
M1	Water pump	1	5V - 12V (to be defined)
PSU1	Power supply	1	Depending on the LED driver
F1	Fuse (optional)	1	Adapted to power supply